Tutorial_9 Answer Key

1. $V_0 = -\frac{R}{L}\int V_1\, dt$, Integration function is being performed.

2. (a) $V_x = V_y = V_b \frac{R_2}{R_1+R_2}$ and $V_0 = \frac{R_2}{R_1}(V_b - V_a)$

   (b) Voltage subtraction
   (c) $V_0 = 3 + 2sin\omega t$

3. Slew will not occur.

4. $V_0 = -6\, V$

5. $V_0 = 3.1\, V\ (peak\ to\ peak)$

6. 2.5 V/μs

7. (a) $V_0 = 8\, V \quad I_0 = 1.47\, mA$

   (b) $\frac{8}{3} < v_a < \frac{32}{3}\ (volts)$

8. $V_0 = -8\, V$, $I_0 = -2\, mA$

9. $V_0 = 8\, V$

10. $V_0 = -(83.3\ sin2t + 1.25t^2)\ mV$